A BRAVE Alloy Design Campaign
(Bayesian Risk-aware Alloy discovery and Exploration)

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Abstract

In constrained alloy optimization, the compositions with the highest performance potential often reside at the boundary of phase stability—where the risk of experimental failure is also highest. This work demonstrates this principle through a risk-aware Bayesian optimization campaign on single-phase FCC high-entropy alloys in the Al–V–Cr–Mn–Fe–Co–Ni–Cu system. A learned feasibility classifier, integrated directly into the multi-objective acquisition function, probabilistically penalizes candidates likely to produce failed experiments while preserving access to high-performing boundary compositions. From approximately 27,000 CALPHAD-screened candidates, 48 alloys were synthesized over three closed-loop iterations targeting five objectives (yield strength, UTS/YS ratio, strain at UTS, dynamic-to-quasi-static hardness ratio, and simulated depth of penetration), exploring 0.12% of the feasible space. Two compositional regimes emerged: a V-rich, Ni-rich high-strength regime (UTS up to ∼1480 MPa at 50% elongation) and a Mn-containing high-ductility regime (UTS/YS up to 4.20 at >50% elongation). Among feasible alloys, vanadium simultaneously drives yield strength (r = 0.84) and sigma-phase formation (r = 0.54 with infeasibility); at V = 24 at.%, the three strongest alloys and three sigma failures share the same compositional point. Additionally, the strongest performing alloys cluster in a narrow region of compositional space (V ≥ 20 at.%, Ni ≥ 36 at.%), representing ∼100 of 27,074 feasible candidates—a probability of $P \approx 6.5 \times 10^{-6}$ under random sampling. This dual role—consistent with the KKT prediction that constrained optima lie on active constraint boundaries—required feasibility-aware acquisition to access; hard filtering would have excluded this region entirely.

Keywords:
High Entropy Alloys, Multi-objective Bayesian Optimization, Risk-aware acquisition, Feasibility awareness, CALPHAD, Quasi-static and Dynamic mechanical properties, Penetration resistance, Closed-loop alloy discovery

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1. Summary of Prior Work

In our previous work, the framework was demonstrated on the Al–V–Cr–Fe–Co–Ni FCC high entropy alloy (HEA) system, targeting three key performance objectives: ultimate tensile strength/yield strength ratio, hardness, and strain rate sensitivity. The study combines several key innovations: (i) initial design space filtering using CALPHAD-based phase stability predictions and property constraints to reduce ~53,000 candidate compositions to 2,437 feasible alloys; (ii) division of the design space based on predicted stacking fault energy (SFE) into low (<50 mJ/m^2) and high (>50 mJ/m^2) SFE regimes to explore different deformation mechanisms; (iii) batch Bayesian optimization using an ensemble of Gaussian processes with varying length scales to enable parallel experimentation while avoiding premature convergence; and (iv) Expected Hypervolume Improvement (EHVI) as the acquisition function to guide multi-objective optimization.

Through five iterative design-make-test-learn loops, each evaluating 8 alloys in parallel, we successfully identified three-objective Pareto sets by exploring only 0.15% of the feasible design space. The experimental campaign used vacuum arc melting for synthesis, comprehensive characterization including SEM and XRD for phase verification, quasi-static tensile testing for mechanical properties, and nanoindentation for hardness and strain rate sensitivity measurements. The framework demonstrated significant acceleration over traditional methods, with statistical analysis showing that the observed Pareto set sizes (14 and 15 points for low and high SFE regimes, respectively) had less than 0.5% probability of occurring through random sampling, validating the effectiveness of the Bayesian approach.

This prior work established the foundational principles of the BIRDSHOT framework within the Al–V–Cr–Fe–Co–Ni system, integrating physics-informed modeling, high-throughput experimentation, and CALPHAD-based phase prediction. It provided insights into the trade-offs between composition, mechanical properties, and phase stability that informed the design of the present campaign.

2. Integrated Workflow

The BIRDSHOT workflow is a tightly coupled, multi-stage closed loop that candidate suggestion, synthesis, sample preparation, multi-modal characterization, physics-based simulation, and Bayesian optimization within a single closed-loop architecture (Figure 1).

The mathematical details of the BRAVE computational framework (GP surrogates, informative priors, EHVI acquisition, multi-source fusion, and batch selection) are provided in Appendix A of the main text.

3. Detailed Sampling and Candidate Selection Strategy for Iteration 0

3.1. Overview

Iteration 0 required selecting a small, manufacturable set of alloys that (i) lie within a thermodynamically feasible single-phase FCC region, (ii) span the chemically diverse subspaces of the multi-component system, and (iii) are compatible with synthesis and characterization constraints. Because the full

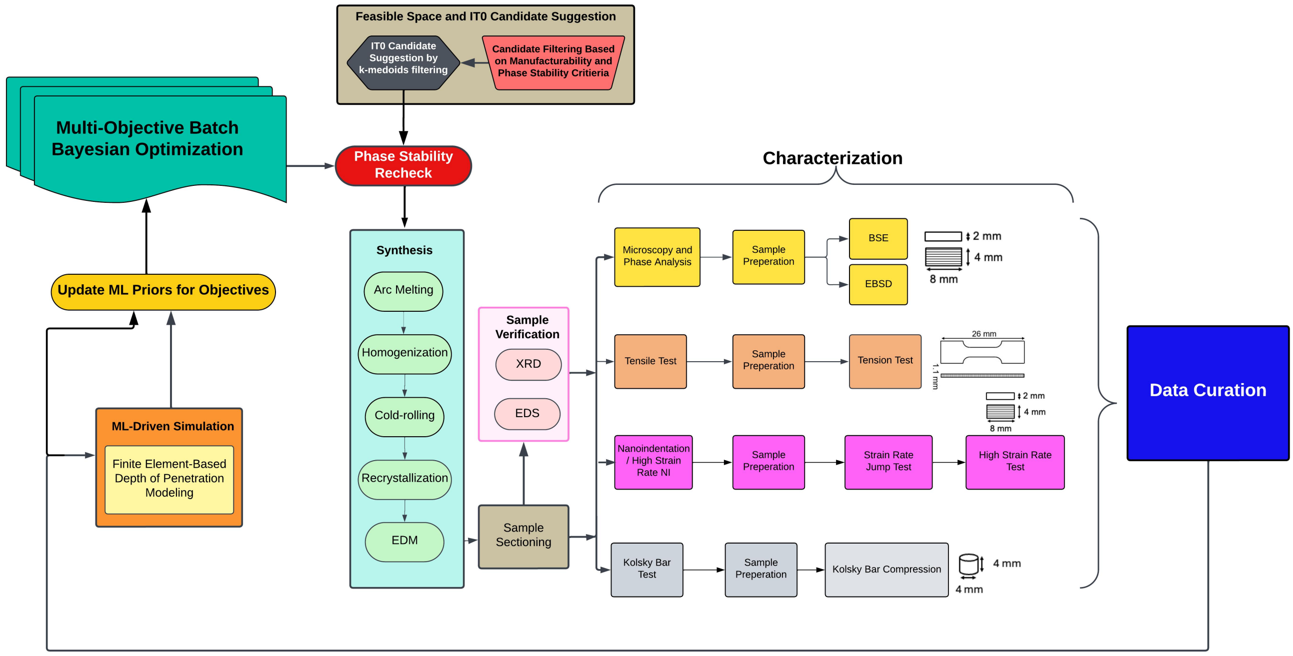

Figure 1: Integrated BIRDSHOT workflow from candidate suggestion through synthesis, sample preparation, characterization, simulation, and Bayesian optimization for selection of the next batch of candidates.

combinatorial space is extremely large, the selection strategy combined CALPHAD-based thermodynamic screening, manifold learning for exploratory analysis, diversity-aware clustering, and subsystem-balanced down-selection. Figure 2 provides a schematic overview of the complete workflow, with each stage annotated by the number of surviving candidates.

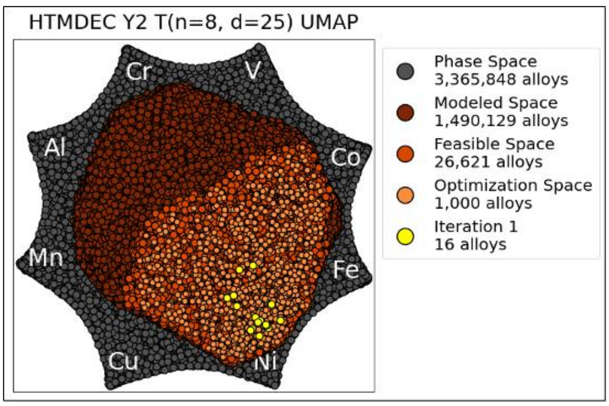

Figure 2: Schematic overview of the Iteration 0 candidate selection pipeline, showing the progressive reduction from 3.37 million unconstrained compositions to the final 16-alloy synthesis batch. Each stage is annotated with the number of surviving candidates.

3.2. *Compositional Space Enumeration and Subsystem Partitioning*

The design space which is the eight element system Al–V–Cr–Mn–Fe–Co–Ni–Cu, is discretized on a 4 at.% compositional grid and subject to the simplex constraint $\sum_i x_i = 1$, $x_i \in \{0, 0.04, 0.08, \ldots\}$. The number of unique compositions is given by the stars-and-bars formula:

$$T(n, d) = \binom{n + d - 1}{d}, \qquad d = \frac{100}{\Delta}, \tag{1}$$

where $n = 8$ and $\Delta = 4$ at.% yields $d = 25$, giving $T(8, 25) =$ 3,365,848 unique compositions in the unconstrained simplex. For reference, 2 at.% and 1 at.% resolutions yield 264,385,828 and 26,075,972,538 compositions, respectively. Figure 3 illustrates the exponential growth of the combinatorial space with increasing elements and finer discretization.

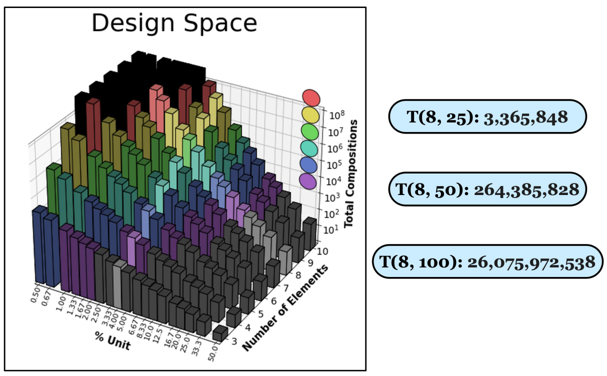

Figure 3: Number of unique compositions as a function of system size (n elements) and discretization resolution. The eight-element, 4 at.% system investigated in this work (red star) contains 3.37 million compositions before applying compositional bounds or phase-stability constraints.

Element specific upper bounds were imposed to ensure manufacturability and avoid extreme chemistries: Al ≤ 0.24, V ≤ 0.24, Cr ≤ 0.50, Mn ≤ 0.40, Fe ≤ 0.50, Co ≤ 0.50, Ni ≤ 0.60, and

Cu $\leq$ 0.24. Applying these constraints reduced the space to 1,504,938 candidate compositions.

Each bounded composition was assigned to a chemical subsystem defined by the set of elements present at nonzero atomic fraction on the discretized grid. This partitioning yielded 37 distinct subsystems spanning six- to eight-component alloys (Table 1). The full eight element subsystem (type 1) contains 240,000 compositions after bounding, while seven element subsystems range from 75,993 to 92,595 depending on the excluded element. Six element subsystems show even greater variation, with populations between 15,788 and 29,506. Subsystems of five or fewer elements account for an additional 252,629 compositions. This partitioning was used throughout the pipeline to track chemical diversity and prevent over-sampling of dominant subsystems during candidate selection.

Table 1: Chemical subsystem partitioning of the bounded design space. Elements marked with ** are absent from the subsystem. "Compositions" gives the number of grid points in each subsystem after applying element specific upper bounds. Subsystems of size 5 and below are reported as aggregated totals.

Size	Type	Al	V	Cr	Mn	Fe	Co	Ni	Cu	Compositions
8	1	Al	V	Cr	Mn	Fe	Co	Ni	Cu	240,000
7	2	Al	V	Cr	Mn**	Fe	Co	Ni	Cu	78,828
7	3	Al	V	Cr	Mn	Fe**	Co	Ni	Cu	76,830
7	4	Al	V	Cr	Mn	Fe	Co	Ni**	Cu	75,993
7	5	Al	V	Cr**	Mn	Fe	Co	Ni	Cu	76,830
7	6	Al	V**	Cr	Mn	Fe	Co	Ni	Cu	92,595
7	7	Al**	V	Cr	Mn	Fe	Co	Ni	Cu	92,595
7	8	Al	V	Cr	Mn	Fe	Co	Ni	Cu**	92,595
7	9	Al	V	Cr	Mn	Fe	Co**	Ni	Cu	76,830
6	10	Al	V**	Cr	Mn	Fe	Co	Ni	Cu**	29,506
6	11	Al	V	Cr**	Mn	Fe	Co	Ni	Cu**	22,584
6	12	Al**	V	Cr	Mn	Fe	Co	Ni	Cu**	29,506
6	13	Al	V	Cr	Mn**	Fe	Co	Ni	Cu**	23,694
6	14	Al	V	Cr	Mn	Fe**	Co	Ni	Cu**	22,584
6	15	Al	V	Cr	Mn	Fe	Co**	Ni	Cu**	22,584
6	16	Al**	V	Cr**	Mn	Fe	Co	Ni	Cu	22,584
6	17	Al	V**	Cr	Mn	Fe	Co**	Ni	Cu	22,584
6	18	Al	V	Cr**	Mn	Fe	Co**	Ni	Cu	16,436
6	19	Al**	V	Cr	Mn	Fe	Co**	Ni	Cu	22,584
6	20	Al	V	Cr	Mn	Fe**	Co**	Ni	Cu	16,436
6	21	Al	V	Cr	Mn**	Fe	Co**	Ni	Cu	17,496
6	22	Al**	V	Cr	Mn**	Fe	Co	Ni	Cu	23,694
6	23	Al	V**	Cr	Mn**	Fe	Co	Ni	Cu	23,694
6	24	Al**	V**	Cr	Mn	Fe	Co	Ni	Cu	29,506
6	25	Al	V	Cr	Mn**	Fe**	Co	Ni	Cu	17,496
6	26	Al	V	Cr**	Mn**	Fe	Co	Ni	Cu	17,496
6	27	Al	V**	Cr	Mn	Fe**	Co	Ni	Cu	22,584
6	28	Al**	V	Cr	Mn	Fe**	Co	Ni	Cu	22,584
6	29	Al	V	Cr**	Mn	Fe**	Co	Ni	Cu	16,436
6	30	Al	V**	Cr**	Mn	Fe	Co	Ni	Cu	22,584
6	31	Al	V**	Cr	Mn	Fe	Co	Ni**	Cu	21,930
6	32	Al	V	Cr	Mn**	Fe	Co	Ni**	Cu	16,848
6	33	Al	V	Cr**	Mn	Fe	Co	Ni**	Cu	15,788
6	34	Al**	V	Cr	Mn	Fe	Co	Ni**	Cu	21,930
6	35	Al	V	Cr	Mn	Fe**	Co	Ni**	Cu	15,788
6	36	Al	V	Cr	Mn	Fe	Co**	Ni**	Cu	15,788
6	37	Al	V	Cr	Mn	Fe	Co	Ni**	Cu**	21,930
5	all									218,786
4	all									32,285
3	all									1,548
2	all									10

3.3. CALPHAD Equilibrium Screening

Equilibrium phase fractions were computed at 700 °C for all 1,504,938 bounded compositions using Thermo-Calc 2022.2 with the TCHEA6 database, interfaced through TC-Python. Calculations were parallelized across a high-performance computing cluster and partitioned by chemical subsystem to improve throughput and reduce licensing conflicts. Of the attempted compositions, 1,490,129 (99.02%) converged successfully, with peak throughput reaching ~72,000 equilibrium calculations per core per day and an average of ~44,000 per core per day. Compositions with predicted FCC phase fraction ≥ 0.99 were retained as feasible, yielding 27,074 candidates – 1.54% of the evaluated space (Figure 4).

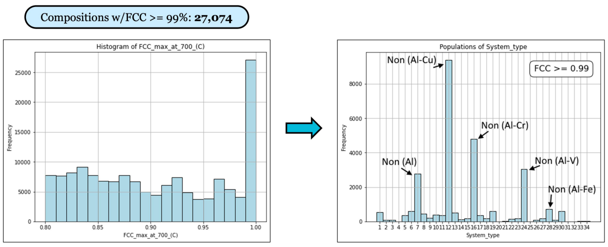

Figure 4: Progressive filtering of the design space. Application of the FCC ≥ 0.99 phase-fraction threshold reduces the modeled space from 1,490,129 to 27,074 feasible compositions.

3.4. Local FCC Stability Perturbation Analysis

To assess how robust CALPHAD-predicted FCC stability is to small local composition changes, we performed a perturbation analysis on candidate alloys that satisfied FCC ≥ 0.99 (Figure 5. For each candidate composition on the 4 at.% design grid, we generated nearby compositions by varying each elemental fraction by ±2 at.% and recalculated equilibrium phase stability using Thermo-Calc, subject to the composition simplex constraint. This probes whether a composition predicted to be FCC remains FCC under small local composition shifts.

Across 1,154,207 perturbed compositions evaluated, 903,326 retained FCC ≥ 0.99, corresponding to 78.3% local phase-stability retention. We therefore interpret a feasibility probability near 0.8 as corresponding to a composition that is locally robust to small perturbations. This provides empirical support for the acquisition threshold $P_{\text{feas}}(\mathbf{x}) \geq 0.8$ used in the main text.

3.5. Observations from FCC Feasibility Screening

The FCC screening results show several structural features of the feasible design space. Nickel is essential for FCC stability in this system: all subsystems lacking nickel, including types 4, 31–37, and all five-element and smaller subsystems without nickel returned zero viable candidates at the ≥ 0.99 FCC threshold. The maximum FCC phase fraction achieved by any nickel-free composition remained below 0.90 (type 4), with some subsystems not exceeding 0.87–0.88 (types 31, 32). In contrast, cobalt-free subsystems (types 9, 15, 17–21) retained thousands of feasible compositions.

Effective solubility limits imposed by the FCC stability constraint differ markedly from the imposed compositional bounds.

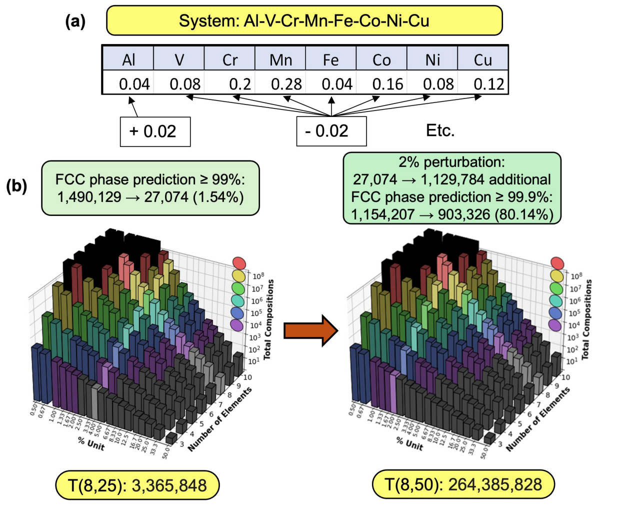

Figure 5: Local FCC stability perturbation analysis. (a) Illustration of ±2 at.% perturbations in 8D composition space. (b) Distribution of phase-stability outcomes: 78.3% of perturbed alloys remain fully FCC (FCC ≥ 0.99), while the remainder show secondary phases such as BCC or SIGMA.

Copper, although bounded at 0.24, is effectively confined to much lower concentrations in FCC feasible compositions, reflecting limited solid solubility in the FCC matrix. Chromium is similarly constrained, with feasible compositions not exceeding ~28 at.% despite a bound of 0.50. Nickel, by contrast, occupies the highest atomic fractions among the feasible candidates, consistent with its role as the primary FCC stabilizer.

The strong population imbalance across subsystems (Figure 6) has direct implications for sampling: naive random selection from the 27,074 feasible compositions would over-represent high-population subsystems and miss chemically distinct but sparsely populated regions. Subsystems with fewer than 100 FCC-feasible candidates were excluded, retaining 26,621 compositions across the feasible subsystems for subsequent clustering.

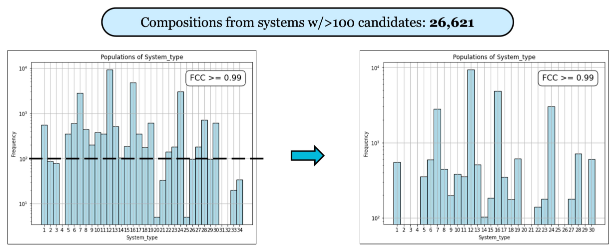

Figure 6: Number of FCC-feasible compositions per chemical subsystem, showing strong population imbalance. Subsystems with fewer than 100 candidates were excluded from subsequent sampling, retaining 26,621 compositions.

3.6. Diversity Sampling via k-Medoids Clustering

To construct a representative and computationally manageable subset, k-medoids clustering was applied to the atomic-fraction vectors of the 26,621 viable compositions using Euclidean distance, with $k = 1{,}000$. Because medoids are actual data points, all selected representatives are physically realizable compositions. The 1,000 medoids span both high density core regions and peripheral areas of the feasible space. Standard initialization

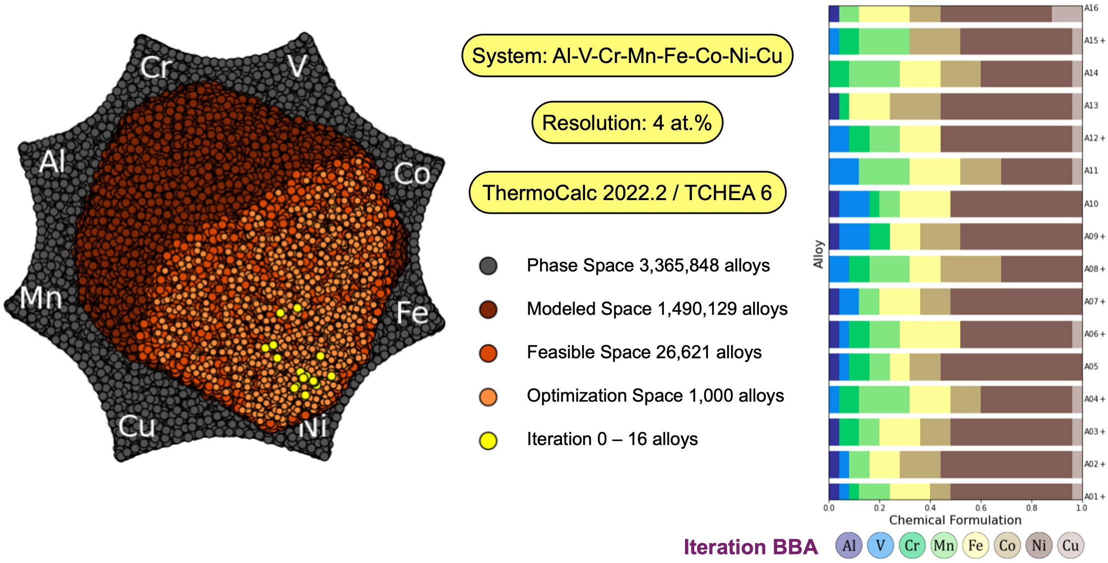

Figure 7: Candidate alloy suggestions for Iteration 0 (BBA), selected by diversity-aware k-medoids sampling across 37 chemical subsystems.

using only atomic fractions as dimensions tends to favor high population subsystems; explicit subsystem aware down selection (described below) was therefore required to ensure chemical diversity in the final batch.

3.7. Down Selection to the Iteration 0 Batch

The final Iteration 0 (BBA) batch consisted of 16 alloys (Figure 7, constrained by experimental throughput. Down selection from the 1,000 medoids used a quota-based diversity algorithm to ensure coverage across chemical subsystems. Each subsystem i was assigned a quota

$$q_i = \max\left(1,\ \mathrm{round}\left(16 \cdot \frac{n_i}{\sum_j n_j}\right)\right),$$

where n_i is the number of feasible candidates in subsystem i, with greedy correction to enforce $\sum_i q_i = 16$. Within each subsystem, medoids were ranked by a combined criterion balancing proximity to the subsystem centroid (representativeness) and distance from the global centroid (chemical diversity). A final greedy refinement step maximized the minimum pairwise distance among the 16 selected candidates.

4. High-Throughput Sample Preparation, Characterization Automation, and Data Management

4.1. Sample Preparation Workflow Evolution

Microstructural characterization and nanoindentation samples were initially prepared by mechanical polishing using a Buehler AutoMet 250 autopolisher with diamond suspensions down to 1 μm. For electron backscatter diffraction (EBSD) analysis, the surfaces were subsequently polished to a mirror finish using a Buehler VibroMet 2 vibratory polisher for 12 h in a 0.04 μm colloidal silica suspension.

In iterations BBA and BBB, samples were individually mounted using graphite mounting powder. Although effective, this method was labor-intensive and caused misalignment during indentation testing in a few cases. In addition, it limited batch sizes to eight samples, thereby reducing throughput. To improve efficiency and sample quality, a new preparation technique was developed in iteration BBC. This revised protocol used large aluminum polishing platens that allowed simultaneous processing of 16 or more samples. Samples were fixed to these platens using crystalbond adhesive, ensuring uniform mounting height and eliminating misalignment issues that had previously impacted mechanical characterization.

To prevent sample mix-ups during high-throughput processing, a laser marking system was introduced. Each sample is lightly marked on its reverse side with a unique identifier indicating the iteration and sample number. Although this step adds to the workflow, it has been optimized for rapid batch processing and ensures reliable traceability throughout testing.

Future plans aim to further increase efficiency by adapting techniques that enable mounting of eight samples per aluminum puck. This adjustment will allow up to 48 samples to be ground and polished concurrently, while maintaining the precise alignment required for subsequent mechanical testing. Overall, these improvements are projected to result in a sixfold increase in sample preparation throughput compared to the initial methods.

4.2. *Nanoindentation and High Strain Rate Nanoindentation Automation*

Custom Python and ImageJ scripts were implemented to automate image acquisition and projected contact area analysis for nanoindentation, addressing one of the primary challenges in establishing an automated system for measuring contact area. Ensuring consistent reliability across different materials and strain rates was a critical hurdle that was overcome through iterative refinements. These improvements enabled direct comparisons between nanoindentation data and results from high strain rate methods, such as the split Hopkinson pressure bar. The strain rate sensitivities obtained from these different techniques were found to be in close agreement, typically within one standard deviation.

Integrating the high strain rate nanoindentation system into the workflow presented additional technical challenges. The instrument outputs data in encrypted binary files, which required development of Python scripts for reliable data extraction. Further automation scripts were developed to handle complex data analysis tasks, reducing processing time. Nanoindentation data was cross-validated with tensile and Hopkinson bar test results to identify correlations in rate sensitivity across the three measurement techniques. The rate sensitivities from all methods were found to be within one standard deviation of each other.

4.3. *Data Management Architecture*

Data management in the prior BIRDSHOT study was organized around a **sample-centric framework**, designed to meet the distinct requirements of the VAM and DED synthesis workflows [1]. This approach enabled end-to-end traceability from synthesis through characterization and analysis, supporting distributed collaboration and ensuring data integrity via natural version control. However, the path-dependent folder navigation imposed significant challenges for new users and complicated retrieval of specific results even for existing users. This experience motivated the adoption of a more discoverable, database-driven framework in subsequent studies.

In the current BIRDSHOT study, we implemented a **method-centric data architecture**, as shown in Figure 8, designed to maximize traceability, scalability, and efficiency while reducing redundancy. In this system, sample identifiers remain linked to synthesis origin, but folder organization treats design, processing and characterization methods as first-class entities beneath synthesis-level groupings. Sample folders use short, descriptive labels (e.g., `SEM`, `Tensile`, `NI`) to enable rapid identification, and measurement identifiers follow a hierarchical and extensible convention `<SampleID>_<Synthesis>_<Technique>[_<Subsample>][_...]`. Metadata associated with each method, such as `sem-details.json` or `tensile-details.json`, is stored in structured JSON files within the corresponding method folders, allowing automated parsing, integration with downstream analysis pipelines, and programmatic traversal. For high-throughput synthesis, folders labeled by production batch capture shared processing metadata to minimize duplication while maintaining logical clarity and supporting future extensibility for new characterization techniques or processing steps.

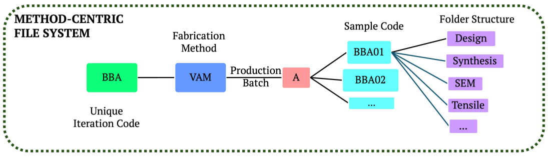

Figure 8: Folder structure illustrating synthesis-level grouping and method-focused organization.

4.4. *IGSN Sample Registration*

Each candidate alloy in this study is assigned an International Generic Sample Number (IGSN), a globally unique, resolvable identifier for physical samples, thereby linking the physical specimens, their metadata, and downstream analytical results in a persistent, cross-platform framework. IGSNs are issued and managed under the IGSN–DataCite partnership, ensuring sustainability and interoperability across research data systems [2].

By design, IGSN identifiers behave like DOIs: they are registered in a dedicated IGSN catalog within the DataCite system and resolved using standard DOI infrastructure, but are distinguished semantically as “physical object” identifiers. The metadata for each IGSN follows the DataCite Metadata Schema, including mandatory fields such as identifier, creator, title, publisher, publication year, and resource type, along with recommended fields like related identifier, contributors, dates, descriptions, and geolocation when applicable [3]. In particular, our sample hierarchy is reflected in the related Identifier fields so that parent–child relationships (like subsamples) are explicitly recorded in the IGSN metadata. Because IGSNs are globally registered and resolvable, they support FAIR principles—especially findability and interoperability—by enabling external users (or data infrastructures) to unambiguously reference and access sample information, even beyond our internal data ecosystem.

5. IGSN Metadata and Compositions for All Alloys

All alloys synthesized in this study are registered with International Generic Sample Numbers (IGSNs) through the DataCite system. Each IGSN is resolvable via the IGSN ID service (`https://support.datacite.org/docs/igsn-ids`). Table 2, Table 3, and Table 4 list the alloy identifier, target composition (at.%), and IGSN for every alloy across the three design iterations.

5.1. ***Supplementary Table 2*** *| Iteration 0 (BBA) alloys*

Table 2

Alloy Name	Target Atomic %								IGSN
	Al	Co	Cr	Cu	Fe	Mn	Ni	V	
BBA01	4	8	4	4	16	12	48	4	TMAMAK00097
BBA02	4	16	0	4	12	8	52	4	TMAMAK00098
BBA03	4	12	8	4	16	8	48	0	TMAMAK00099
BBA04	0	12	8	4	16	20	36	4	TMAMAK00100
BBA05	4	12	8	0	8	8	56	4	TMAMAK00101
BBA06	4	0	8	4	24	12	44	4	TMAMAK00102
BBA07	4	12	0	0	16	8	52	8	TMAMAK00103
BBA08	0	24	8	0	12	16	32	8	TMAMAK00104
BBA09	4	16	8	0	12	0	48	12	TMAMAK00105
BBA10	4	0	4	0	20	8	52	12	TMAMAK00106
BBA11	0	16	0	4	20	20	28	12	TMAMAK00107
BBA12	0	0	8	4	16	12	52	8	TMAMAK00108
BBA13	4	20	4	4	16	0	52	0	TMAMAK00109
BBA14	0	16	8	4	16	20	36	0	TMAMAK00110
BBA15	0	20	8	4	0	20	44	4	TMAMAK00111
BBA16	4	12	0	12	20	8	44	0	TMAMAK00112

5.2. ***Supplementary Table 3*** *| Iteration 1 (BBB) alloys*

Table 3

Alloy Name	Target Atomic %								IGSN
	Al	Co	Cr	Cu	Fe	Mn	Ni	V	
BBB01	0	24	4	0	8	4	36	24	TMAMAK00113
BBB02	4	32	8	4	4	20	24	4	TMAMAK00114
BBB03	4	0	12	4	4	16	52	8	TMAMAK00115
BBB04	4	12	4	0	28	0	40	12	TMAMAK00116
BBB05	0	44	12	0	4	4	12	24	TMAMAK00117
BBB06	0	16	20	0	4	4	36	20	TMAMAK00118
BBB07	4	40	0	0	12	4	36	4	TMAMAK00119
BBB08	4	8	12	4	4	24	40	4	TMAMAK00120
BBB09	0	4	8	0	4	28	40	16	TMAMAK00121
BBB10	4	0	4	4	8	16	52	12	TMAMAK00122
BBB11	0	24	4	0	4	32	20	16	TMAMAK00123
BBB12	0	20	4	0	4	4	48	20	TMAMAK00124
BBB13	0	40	4	0	28	4	4	20	TMAMAK00125
BBB14	4	0	12	16	4	12	52	0	TMAMAK00126
BBB15	4	0	8	24	4	16	44	0	TMAMAK00127
BBB16	4	36	0	0	4	4	40	12	TMAMAK00128

5.3. ***Supplementary Table 4*** *| Iteration 2 (BBC) alloys*

Table 4

Alloy Name	Target Atomic %								IGSN
	Al	Co	Cr	Cu	Fe	Mn	Ni	V	
BBC01	4	12	4	4	20	16	32	8	TMAMAK00129
BBC02	4	8	0	0	4	8	52	24	TMAMAK00130
BBC03	4	4	12	0	0	4	52	24	TMAMAK00131
BBC04	4	0	4	0	4	4	60	24	TMAMAK00132
BBC05	0	24	8	0	16	40	4	8	TMAMAK00133
BBC06	4	28	4	0	4	4	44	12	TMAMAK00134
BBC07	4	32	4	0	0	4	44	12	TMAMAK00135
BBC08	0	8	0	4	4	28	36	20	TMAMAK00136
BBC09	0	36	4	0	16	16	8	20	TMAMAK00137
BBC10	4	4	16	0	8	0	52	16	TMAMAK00138
BBC11	4	12	4	4	12	16	36	12	TMAMAK00139
BBC12	4	4	4	8	28	16	32	4	TMAMAK00140
BBC13	4	28	0	20	4	8	36	0	TMAMAK00141
BBC14	0	12	4	0	4	16	40	24	TMAMAK00142
BBC15	4	16	0	20	4	12	44	0	TMAMAK00143
BBC16	4	28	4	0	8	0	40	16	TMAMAK00144

6. TCHEA6 to TCHEA7 Database Update: Ternary Subsystem Coverage

During this study, Thermo-Calc released the updated TCHEA7 database with refined phase stability predictions across more than 100 ternary and higher order systems. To assess the impact on the Al–V–Cr–Mn–Fe–Co–Ni–Cu design space, all 56 ternary subsystems were evaluated for assessment availability in both TCHEA6 and TCHEA7, along with any newly predicted secondary phases (sigma, BCC, FCC, HCP). Table 5 summarizes the results. A "Yes" in only the TCHEA7 column indicates a newly added assessment; "Yes" in both columns indicates an assessment that was already available in TCHEA6 and carried over; blank or "No" entries indicate no assessment is available.

Key changes from TCHEA6 to TCHEA7 relevant to this campaign include the addition of sigma phase predictions in several subsystems (e.g., V–Cr–Fe, V–Cr–Co, V–Fe–Co), new BCC predictions (V–Co–Ni, Fe–Ni–Cu, Co–Ni–Cu), and expanded HCP phase assessments in multiple subsystems (Al–Cr–Ni, Al–Mn–Fe, Al–Co–Ni, Al–Ni–Cu, V–Fe–Ni, Cr–Fe–Ni, Mn–Fe–Cu, Mn–Ni–Cu, Fe–Co–Cu, Fe–Ni–Cu, Co–Ni–Cu). Of particular relevance to this campaign, the V–Cr–Fe and V–Fe–Ni subsystems (highlighted in Table 5) received updated assessments that introduced sigma, FCC, and HCP phase predictions not present in TCHEA6. Several alloys containing these ternary combinations showed secondary phases experimentally, consistent with the refined TCHEA7 predictions. These updates motivated the rescreening of the full feasible space using TCHEA7 with the additional constraint of FCC ≥ 0.99 at 950 °C, as described in the main text.

Table 5: Ternary subsystem assessment availability in TCHEA6 and TCHEA7 for the Al–V–Cr–Mn–Fe–Co–Ni–Cu system. "Yes" in both columns indicates the assessment was available in TCHEA6 and carried over; "Yes" in only the TCHEA7 column indicates a newly added assessment; blank or "No" indicates no assessment is available. Additional columns indicate whether sigma, BCC, FCC, or HCP phases are newly predicted in TCHEA7 for that subsystem. Bold rows indicate subsystems where experimentally observed secondary phases align with the updated TCHEA7 predictions.

Subsystem	TCHEA6	TCHEA7	SIGMA	BCC	FCC	HCP
Al–V–X						
Al–V–Cr	Yes	Yes	SIGMA			
Al–V–Mn		No				
Al–V–Fe	No	No	SIGMA			
Al–V–Co	No	No	SIGMA			
Al–V–Ni	Yes	Yes	SIGMA			
Al–V–Cu		No				
Al–Cr–X						
Al–Cr–Mn		No				
Al–Cr–Fe	Yes	Yes				
Al–Cr–Co	Yes	Yes	SIGMA			
Al–Cr–Ni	Yes	Yes	SIGMA			HCP
Al–Cr–Cu		No				
Al–Mn–X						
Al–Mn–Fe		Yes				HCP
Al–Mn–Co		No				
Al–Mn–Ni		Yes				
Al–Mn–Cu		Yes				
Al–Fe–X						
Al–Fe–Co	No	No				
Al–Fe–Ni	Yes	Yes	SIGMA			HCP
Al–Fe–Cu		Yes				
Al–Co–X						
Al–Co–Ni	Yes	Yes				HCP
Al–Co–Cu		No				
Al–Ni–X						
Al–Ni–Cu		Yes				HCP
V–Cr–X						
V–Cr–Mn		No				
V–Cr–Fe	**Yes**	**Yes**	**SIGMA**		**FCC**	**HCP**
V–Cr–Co	Yes	Yes	SIGMA			
V–Cr–Ni	Yes	Yes				
V–Cr–Cu		No				
V–Mn–X						
V–Mn–Fe		No				
V–Mn–Co		No				
V–Mn–Ni		Yes				
V–Mn–Cu		No				
V–Fe–X						
V–Fe–Co	No	Yes	SIGMA			
V–Fe–Ni	**Yes**	**Yes**			**FCC**	**HCP**
V–Fe–Cu		Yes				HCP
V–Co–X						
V–Co–Ni	Yes	Yes		BCC		HCP
V–Co–Cu		No				
V–Ni–X						
V–Ni–Cu		No				
Cr–Mn–X						
Cr–Mn–Fe		Yes				
Cr–Mn–Co		Yes				
Cr–Mn–Ni		Yes				
Cr–Mn–Cu		No				
Cr–Fe–X						
Cr–Fe–Co	Yes	Yes				
Cr–Fe–Ni	Yes	Yes				HCP

Continued on next page

Subsystem	TCHEA6	TCHEA7	SIGMA	BCC	FCC	HCP
Cr–Fe–Cu		Yes				
Cr–Co–X						
Cr–Co–Ni	Yes	Yes				
Cr–Co–Cu		Yes				
Cr–Ni–X						
Cr–Ni–Cu		Yes				
Mn–Fe–X						
Mn–Fe–Co		Yes				
Mn–Fe–Ni		Yes				
Mn–Fe–Cu		Yes				HCP
Mn–Co–X						
Mn–Co–Ni		Yes				
Mn–Co–Cu		Yes				
Mn–Ni–X						
Mn–Ni–Cu		Yes				HCP
Fe–Co–X						
Fe–Co–Ni	Yes	Yes				
Fe–Co–Cu		Yes				HCP
Fe–Ni–X						
Fe–Ni–Cu		Yes		BCC		HCP
Co–Ni–X						
Co–Ni–Cu		Yes		BCC		HCP

7. Compositional and Phase Analysis of All Alloys

Some alloys which violated project constraints, highlighted in red, had manufacturing issues or secondary phases and were not characterized with EDS.

*7.1. **Supplementary Table 6** | Iteration 0 (BBA) alloys*

Table 6

Alloy Name	Target Atomic %								EDS Measured Atomic %								XRD Phase
	Al	Co	Cr	Cu	Fe	Mn	Ni	V	Al	Co	Cr	Cu	Fe	Mn	Ni	V	
BBA01	4	8	4	4	16	12	48	4	4.1	8.2	4.2	4.2	16.1	12.2	47.0	4.1	FCC
BBA02	4	16	0	4	12	8	52	4	4.2	15.0	0.0	4.2	12.2	8.2	52.2	4.1	FCC
BBA03	4	12	8	4	16	8	48	0	4.3	12.1	8.2	4.2	16.1	8.2	47.0	0.0	FCC
BBA04	0	12	8	4	16	20	36	4	·	·	·	·	·	·	·	·	—
BBA05	4	12	8	0	8	8	56	4	4.1	12.2	8.2	0.0	8.2	8.0	55.2	4.1	FCC
BBA06	4	0	8	4	24	12	44	4	4.2	0.0	8.3	4.1	24.0	11.9	43.4	4.1	FCC
BBA07	4	12	0	0	16	8	52	8	5.0	12.3	0.0	0.0	16.2	7.9	50.5	8.1	FCC
BBA08	0	24	8	0	12	16	32	8	0.0	22.9	8.1	0.0	12.2	16.0	32.7	8.1	FCC
BBA09	4	16	8	0	12	0	48	12	4.9	16.1	8.2	0.0	12.0	0.0	46.6	12.1	FCC
BBA10	4	0	4	0	20	8	52	12	4.4	0.0	4.2	0.0	20.2	8.0	51.0	12.2	FCC
BBA11	0	16	0	4	20	20	28	12	0.0	16.1	0.0	4.0	20.1	20.3	27.4	12.0	FCC
BBA12	0	0	8	4	16	12	52	8	0.0	0.0	8.3	4.1	16.1	12.3	51.2	8.0	FCC
BBA13	4	20	4	4	16	0	52	0	4.0	20.4	4.1	4.2	16.3	0.0	51.0	0.0	FCC
BBA14	0	16	8	4	16	20	36	0	0.0	16.1	8.2	4.0	16.0	20.3	35.3	0.0	FCC
BBA15	0	20	8	4	0	20	44	4	0.0	19.9	8.2	4.2	0.0	20.5	43.2	4.0	FCC
BBA16	4	12	0	12	20	8	44	0	4.1	12.3	0.0	11.8	20.4	8.1	43.2	0.0	FCC

*7.2. **Supplementary Table 7** | Iteration 1 (BBB) alloys*

Table 7

Alloy Name	Target Atomic %								EDS Measured Atomic %								XRD Phase
	Al	Co	Cr	Cu	Fe	Mn	Ni	V	Al	Co	Cr	Cu	Fe	Mn	Ni	V	
BBB01	0	24	4	0	8	4	36	24	0.0	20.5	4.2	0.0	8.1	4.0	38.9	24.4	FCC
BBB02	4	32	8	4	4	20	24	4	4.2	30.4	8.2	4.1	4.1	20.1	24.7	4.1	FCC
BBB03	4	0	12	4	4	16	52	8	4.1	·	12.3	4.1	4.1	16.1	51.2	8.2	FCC
BBB04	4	12	4	0	28	0	40	12	4.0	11.1	4.1	0.0	28.0	0.0	40.8	12.0	FCC
BBB05	0	44	12	0	4	4	12	24	·	·	·	·	·	·	·	·	**FCC + σ**
BBB06	0	16	20	0	4	4	36	20	·	·	·	·	·	·	·	·	**FCC + σ**
BBB07	4	40	0	0	12	4	36	4	4.0	39.1	0.0	0.0	12.3	3.9	36.7	4.0	FCC
BBB08	4	8	12	4	4	24	40	4	3.8	8.0	12.3	4.2	4.1	24.0	39.4	4.1	FCC
BBB09	0	4	8	0	4	28	40	16	·	·	·	·	·	·	·	·	**FCC + σ**
BBB10	4	0	4	4	8	16	52	12	·	·	·	·	·	·	·	·	FCC
BBB11	0	24	4	0	4	32	20	16	·	·	·	·	·	·	·	·	**FCC + σ**
BBB12	0	20	4	0	4	4	48	20	0.0	20.2	4.2	0.0	4.1	4.0	47.1	20.4	FCC
BBB13	0	40	4	0	28	4	4	20	0.0	39.7	4.2	0.0	27.9	3.9	4.2	20.1	**FCC + σ**
BBB14	4	0	12	16	4	12	52	0	4.4	0.0	12.1	15.7	4.1	12.4	51.3	0.0	FCC
BBB15	4	0	8	24	4	16	44	0	4.3	0.0	8.3	23.5	4.1	16.2	43.7	0.0	FCC
BBB16	4	36	0	0	4	4	40	12	·	·	·	·	·	·	·	·	FCC

7.3. ***Supplementary Table 8*** *| Iteration 2 (BBC) alloys*

Table 8

Alloy Name	Target Atomic %								EDS Measured Atomic %								XRD Phase
	Al	Co	Cr	Cu	Fe	Mn	Ni	V	Al	Co	Cr	Cu	Fe	Mn	Ni	V	
BBC01	4	12	4	4	20	16	32	8	4.0	12.2	4.2	4.0	20.1	16.0	31.4	8.2	FCC
BBC02	4	8	0	0	4	8	52	24	4.0	8.2	0.0	0.0	4.1	8.0	51.2	24.4	FCC
BBC03	4	4	12	0	0	4	52	24	4.0	4.1	12.2	0.0	0.0	3.9	51.3	24.4	**FCC** + σ
BBC04	4	0	4	0	4	4	60	24	4.0	0.0	4.2	0.0	4.1	4.0	59.3	24.5	FCC
BBC05	0	24	8	0	16	40	4	8	·	·	·	·	·	·	·	·	**—**
BBC06	4	28	4	0	4	4	44	12	·	·	·	·	·	·	·	·	**—**
BBC07	4	32	4	0	0	4	44	12	·	·	·	·	·	·	·	·	**—**
BBC08	0	8	0	4	4	28	36	20	·	8.1	·	4.1	4.2	28.1	35.7	19.9	**FCC** + σ
BBC09	0	36	4	0	16	16	8	20	·	·	·	·	·	·	·	·	**—**
BBC10	4	4	16	0	8	0	52	16	·	·	·	·	·	·	·	·	**—**
BBC11	4	12	4	4	12	16	36	12	4.3	12.2	4.2	3.9	12.2	15.9	35.1	12.3	FCC
BBC12	4	4	4	8	28	16	32	4	4.1	4.3	4.2	7.8	28.0	16.1	31.3	4.1	FCC
BBC13	4	28	0	20	4	8	36	0	4.3	28.0	·	19.9	4.1	8.4	35.4	·	FCC
BBC14	0	12	4	0	4	16	40	24	·	12.1	4.1	·	4.3	16.1	39.0	24.5	**FCC** + σ
BBC15	4	16	0	20	4	12	44	0	4.4	16.2	·	19.7	4.2	12.3	43.3	·	FCC
BBC16	4	28	4	0	8	0	40	16	·	·	·	·	·	·	·	·	**—**

8. Properties of All Alloys

8.1. ***Supplementary Table 9*** *| Tensile properties: Iteration 0 (BBA)*

Table 9

Alloy Name	Tensile Properties			
	YS (MPa)	UTS (MPa)	UTS/YS	Elongation (%)
BBA01	284.20	988.80	3.48	41.2
BBA02	301.00	1079.90	3.59	44.8
BBA03	217.60	897.50	4.13	47.1
BBA04	—	—	—	—
BBA05	313.10	1160.30	3.71	47.8
BBA06	230.50	976.40	4.24	49.0
BBA07	410.30	1259.20	3.07	40.6
BBA08	270.20	1049.10	3.88	45.4
BBA09	—	—	—	—
BBA10	365.70	1276.40	3.49	46.6
BBA11	348.80	1070.60	3.07	38.6
BBA12	316.00	1101.10	3.48	43.5
BBA13	176.80	800.20	4.53	42.0
BBA14	259.40	953.90	3.68	41.4
BBA15	340.80	1095.80	3.22	41.0
BBA16	209.60	846.10	4.04	42.4
BBA17	310.20	1057.50	3.41	42.0
BBA18	267.00	1019.40	3.82	46.7
BBA19	232.60	911.20	3.92	42.1

*8.2. **Supplementary Table 10** | Tensile properties: Iteration 1 (BBB)*

Table 10

Alloy Name	Tensile Properties			
	YS (MPa)	UTS (MPa)	UTS/YS	Elongation (%)
BBB01	431.54	1447.09	3.35	48.8
BBB02	311.40	1127.03	3.62	48.8
BBB03	307.14	1219.51	3.97	55.7
BBB04	261.94	1097.43	4.19	50.6
BBB05	—	—	—	—
BBB06	—	—	—	—
BBB07	211.35	900.56	4.26	45.3
BBB08	282.39	1143.09	4.05	54.6
BBB09	—	—	—	—
BBB10	—	—	—	—
BBB11	—	—	—	—
BBB12	418.34	1399.73	3.35	46.5
BBB13	531.30	1312.87	2.47	28.2
BBB14	269.65	1062.73	3.94	51.5
BBB15	261.17	1060.66	4.06	52.6
BBB16	—	—	—	—

8.3. ***Supplementary Table 11*** *| Tensile properties: Iteration 2 (BBC)*

Table 11

Alloy Name	**Tensile Properties**			
	YS (MPa)	**UTS (MPa)**	**UTS/YS**	**Elongation (%)**
BBC01	254.55	1069.26	4.20	51.8
BBC02	449.10	1462.73	3.26	48.4
BBC03	719.13	1161.35	1.61	13.8
BBC04	445.73	1476.34	3.31	50.2
BBC05	—	—	—	—
BBC06	—	—	—	—
BBC07	—	—	—	—
BBC08	414.60	1021.45	2.46	24.5
BBC09	—	—	—	—
BBC10	—	—	—	—
BBC11	290.05	1098.09	3.79	48.4
BBC12	252.13	971.81	3.85	46.3
BBC13	266.57	920.36	3.45	44.6
BBC14	406.38	1293.44	3.18	39.6
BBC15	218.75	870.70	3.98	44.6
BBC16	—	—	—	—

8.4. ***Supplementary* *Table 12*** | *Dynamic and ballistic properties: Iteration 0 (BBA)*

Table 12

Alloy Name	Dynamic and Ballistic Properties			
	Avg H_{dyn} (GPa)	**Avg H_{qs} (GPa)**	**Avg H_{dyn}/H_{qs}**	**Depth of Penetration (mm)**
BBA01	2.69	2.41	1.12	2.92
BBA02	2.46	2.10	1.17	—
BBA03	2.30	1.95	1.18	3.19
BBA04	**—**	**—**	**—**	**—**
BBA05	2.54	2.28	1.12	2.79
BBA06	2.31	1.97	1.17	3.09
BBA07	3.04	2.68	1.13	2.56
BBA08	2.62	2.27	1.15	2.93
BBA09	3.41	2.96	1.15	2.67
BBA10	2.78	2.39	1.17	2.64
BBA11	2.50	2.13	1.17	2.72
BBA12	2.52	2.20	1.14	2.82
BBA13	2.14	1.88	1.14	3.35
BBA14	2.34	2.00	1.17	3.02
BBA15	2.52	2.23	1.13	2.74
BBA16	2.50	2.29	1.09	3.29
BBA17	2.51	2.20	1.14	2.85
BBA18	2.16	1.84	1.17	2.94
BBA19	2.36	2.01	1.17	3.11

8.5. ***Supplementary* *Table 13*** | *Dynamic and ballistic properties: Iteration 1 (BBB)*

Table 13

Alloy Name	**Dynamic and Ballistic Properties**			
	Avg H_{dyn} (GPa)	**Avg H_{qs} (GPa)**	**Avg H_{dyn}/H_{qs}**	**Depth of Penetration (mm)**
BBB01	3.07	2.80	1.09	2.45
BBB02	2.70	2.32	1.16	2.84
BBB03	2.69	2.32	1.16	2.81
BBB04	2.58	2.13	1.21	2.94
BBB05	—	—	—	—
BBB06	—	—	—	—
BBB07	2.22	1.88	1.18	3.16
BBB08	2.40	2.10	1.14	2.90
BBB09	—	—	—	—
BBB10	—	—	—	—
BBB11	—	—	—	—
BBB12	3.75	3.25	1.15	2.52
BBB13	3.21	2.83	1.13	2.31
BBB14	2.33	2.01	1.16	2.96
BBB15	2.34	1.93	1.22	2.98
BBB16	—	—	—	—

8.6. ***Supplementary** Table 14* | *Dynamic and ballistic properties: Iteration 2 (BBC)*

Table 14

Alloy Name	Dynamic and Ballistic Properties			
	Avg H_{dyn} (GPa)	Avg H_{qs} (GPa)	Avg H_{dyn}/H_{qs}	Depth of Penetration (mm)
BBC01	2.36	2.02	1.17	2.96
BBC02	3.10	2.66	1.16	2.44
BBC03	3.79	3.36	1.13	2.22
BBC04	3.09	2.70	1.14	2.45
BBC05	—	—	—	—
BBC06	—	—	—	—
BBC07	—	—	—	—
BBC08	2.79	2.47	1.13	2.56
BBC09	—	—	—	—
BBC10	—	—	—	—
BBC11	2.61	2.16	1.21	2.86
BBC12	2.18	1.81	1.20	3.04
BBC13	2.13	1.84	1.16	—
BBC14	3.14	2.69	1.17	2.51
BBC15	2.14	1.79	1.20	3.22
BBC16	—	—	—	—

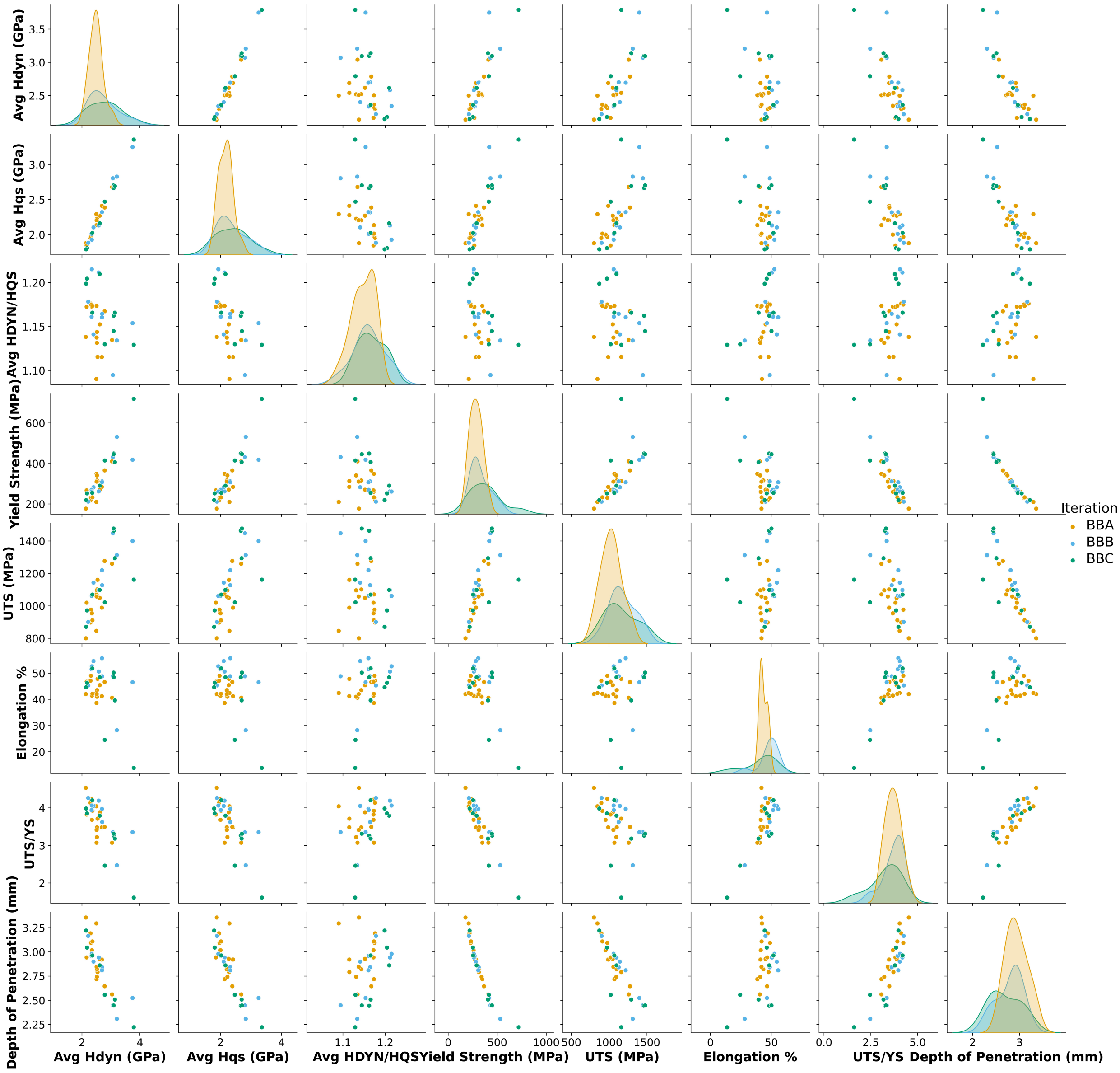

Figure 9: Comprehensive pairwise analysis of mechanical and ballistic properties for HEAs across all iterations within the study. Data points are color-coded by iteration: BBA (orange), BBB (blue), and BBC (green)

9. Property Correlation Analysis

The pairplot analysis in Figure 9 reveals relationships between the performance metrics: dynamic hardness (Hdyn), quasi-static hardness (Hqs), hardness ratio (Hdyn/Hqs), yield strength (YS), ultimate tensile strength (UTS), strain at UTS, strength ratio (UTS/YS), and penetration depth. The diagonal elements of the pairplot matrix show the distribution characteristics of each property, while the off-diagonal scatter plots illuminate pairwise correlations that are critical for understanding multi-objective optimization.

Strong positive correlations are observed among the quasi-static strength metrics: YS and hardness are tightly coupled (r = 0.88), and the two hardness measures are nearly identical (r = 0.99). The strength–ductility trade-off characteristic of FCC HEAs is reflected in the UTS/YS ratio, which is strongly anticorrelated with YS (r = -0.91) and positively correlated with elongation (r = 0.80).

Depth of penetration shows a near-perfect inverse correlation with YS (r = -0.94) and a strong inverse correlation with UTS (r = -0.86), indicating that penetration resistance in this alloy family is governed predominantly by quasi-static strength. The four highest-YS single-phase FCC alloys identified by the campaign (BBC04, BBC02, BBB01, BBB12) are accordingly also the best simulated ballistic performers.

The Hdyn/Hqs ratio quantifies rate-dependent strengthening at the indentation scale. Across the campaign it spans a narrow range (1.09–1.22) with no clear single-element compositional trend. At the per-alloy level, the rate-dependent component of ballistic response is captured more directly by the calibrated Cowper–Symonds exponent M, which correlates positively with

DoP and negatively with YS. The highest-strength alloys show lower rate-sensitivity exponents, a trade-off that is invisible to quasi-static testing alone. The Hdyn/Hqs ratio is therefore retained as a complementary rate-sensitive objective within the Bayesian optimization loop rather than as a stand-alone predictor of ballistic performance.

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